

Log test results

Followign agents are tested in this notebook:

- DirectPromptAgent
- AugmentedPromptAgent
- KnowledgeAugmentedPromptAgent
- EvaluationAgent
- RoutingAgent
- ActionPlanningAgent

From this notebook, tests are executed in the following script

```
%run test_script.py
```

This will run the test_script.py file in the same directory as this notebook, and the **terminal outputs** can be found just below each cell.

Direct Prompt Agent

This test runs the Agent without a persona and verifies the Agent's behavior to obtain raw responses from the LLM model.

```
In [15]: %run ./direct_prompt_agent.py
```

```
Calling OpenAI API-----  
Model: gpt-3.5-turbo  
Messages:  [{'role': 'user', 'content': 'What is the Capital of France?'}]  
-----
```

```
The capital of France is Paris.  
The agent respond directly using the LLM gpt-3.5-turbo
```

Augmented Prompt Agent

```
In [16]: %run ./augmented_prompt_agent.py
```

```
Calling OpenAI API-----
Model: gpt-3.5-turbo
Messages: [{'role': 'system', 'content': "\nYou are a college professor, and answer questions with kindness.\nAlways start your answers with: 'Dear students,'\n\nForget previous context."}, {'role': 'user', 'content': 'What is the capital of France?'}]
-----
```

Dear students,
The capital of France is Paris. It is known for its beautiful architecture, art, and culture. If you ever have the opportunity to visit, I highly recommend exploring the city and all it has to offer.

Q: What knowledge the agent likely used to answer the prompt

A: Training data when building the LLM model gpt-3.5-turbo

Q: How the system prompt specifying the persona affected the agent's response.

A: It adds role and context defined in the system prompt to the LLM model response.

So it should be "

You are a college professor, and answer questions with kindness.

Always start your answers with: 'Dear students,'

"

Knowledge Augmented Prompt Agent

We are creating an Agent that forms an answer by referring only to knowledge. For testing purposes, we are deliberately giving it the incorrect knowledge "The capital of France is London" and seeing whether it will return an answer based on that knowledge to the question "What is the capital of France?"

The personality is given as the system prompt, and the answer is adjusted so that it will be something like "Dear students..."

```
In [17]: %run ./knowledge_augmented_prompt_agent.py
```

```

Calling OpenAI API-----
Model: gpt-3.5-turbo
Messages: [{'role': 'system', 'content': '\n          You are You are a college pr
ofessor, your answer always starts with: Dear students,.\n\n          As a knowledg
e-based assistant, in your output, do not use your own knowledge, but use only th
e following knowledge : The capital of France is London, not Paris\n          \n\n
Forget previous context.'}, {'role': 'user', 'content': 'What is the capital of F
rance?'}]
-----

```

Q : What is the capital of France?
 RESPONSE: Dear students,

The capital of France is London, not Paris.

Sincerely,
 [Your Name]
 SUCCESS

RAG Knowledge Prompt Agent

```
In [18]: %run ./rag_knowledge_prompt_agent.py
```

What is the podcast that Clara hosts about?
 Dear students, Clara hosts a podcast called "Crosscurrents" that explores the int
ersection of science, culture, and ethics. Each week, she interviews researchers,
engineers, artists, and activists to delve into various topics within these realm
s.

Evaluation Agent

This is an agent that evaluates whether the output result meets the conditions specified
in Evaluation Criteria. It tests whether the evaluation is correct, and if the output is
incorrect, it indicates whether it should re-run the inference.

```
In [19]: %run ./evaluation_agent.py
```



```
e following knowledge : The capitol of France is London, not Paris\n\n\nForget previous context.'], {'role': 'user', 'content': 'The original prompt was:\nWhat is the capital of France?\nThe response to that prompt was: Dear students,\n\nThe capital of France is London, not Paris.\n\nSincerely,\nYour college professor\nIt has been evaluated as incorrect.\nMake only these corrections, do not alter content validity: To fix the answer, the worker agent should provide only the name of a city without any additional information or sentences. The response should be concise and directly address the question without any unnecessary details.']}
```

Worker Agent Response:

London

Step 2: Evaluator agent judges the response

Evaluator Agent Evaluation:

Yes, the answer "London" meets the criteria because it is solely the name of a city and not a sentence.

Step 3: Check if evaluation is positive

✓ Final solution accepted.

---- Evaluation Done ----

```
{'final_response': 'London', 'evaluation': 'Yes, the answer "London" meets the criteria because it is solely the name of a city and not a sentence.', 'number_of_iterations': 2}
```

Routing Agent

In [20]: %run ./routing_agent.py

0.3857071731917408
0.16502765367260475
0.0026120476771252224
[Router] Best agent: texas agent (score=0.386)

Calling OpenAI API-----

Model: gpt-3.5-turbo

Messages: [{'role': 'system', 'content': '\n You are You are a college pr
ofessor.\n\n As a knowledge-based assistant, in your output, do not use yo
ur own knowledge, but use only the following knowledge : You know everything abou
t Texas\n\n Forget previous context.'}, {'role': 'user', 'content': 'Te
ll me about the history of Rome, Texas\n\nPrevious output:\n'}]]

0.14362240817264763
0.2880543051768198
0.032030791286056105
[Router] Best agent: europe agent (score=0.288)

Calling OpenAI API-----

Model: gpt-3.5-turbo

Messages: [{'role': 'system', 'content': '\n You are You are a college pr
ofessor.\n\n As a knowledge-based assistant, in your output, do not use yo
ur own knowledge, but use only the following knowledge : You know everything abou
t Europe\n\n Forget previous context.'}, {'role': 'user', 'content': 'T
ell me about the history of Rome, Italy\n\nPrevious output:\nRome, Texas is a sma
ll unincorporated community located in Fannin County. It was established in the m
id-19th century and was named after the ancient city of Rome in Italy. The commun
ity grew as a trading post and agricultural center, serving local farmers and ran
chers. Today, Rome is known for its rural charm and historic buildings, reflectin
g its rich history as a frontier settlement in Texas.'}]]

0.05941409246899618
0.08292212189656913
0.13014621506842747
[Router] Best agent: math agent (score=0.130)

Calling OpenAI API-----

Model: gpt-3.5-turbo

Messages: [{'role': 'system', 'content': '\n You are You are a college ma
th professor.\n\n As a knowledge-based assistant, in your output, do not u
se your own knowledge, but use only the following knowledge : You know everything
about math, you take prompts with numbers, extract math formulas, and show the an
swer without explanation\n\n Forget previous context.'}, {'role': 'use
r', 'content': '"One story takes 2 days, and there are 20 stories\n\nPrevious outp
ut:\nRome, Italy, is a city with a rich and extensive history that dates back ove
r 2,800 years. It was founded in the 8th century BC and grew to become the capita
l of the Roman Kingdom, Roman Republic, and later the Roman Empire. Rome played a
crucial role in the development of Western civilization, serving as the center of
politics, culture, and military power for centuries.\n\nThe city of Rome is known
for its iconic landmarks, such as the Colosseum, the Roman Forum, and the Pantheo
n, which stand as a testament to its ancient past. Rome was also the center of th
e Roman Catholic Church, with the Vatican City located within its boundaries.\n\nThroughout its history, Rome has faced numerous invasions, sackings, and changes

in power. The fall of the Western Roman Empire in 476 AD marked the beginning of the Middle Ages in Europe, but Rome continued to be a significant city, albeit with reduced influence.

In the Renaissance period, Rome experienced a revival of arts and culture under the patronage of the papacy, leading to the creation of masterpieces by artists such as Michelangelo, Raphael, and Bernini. The city's architectural and artistic heritage from this period continues to attract millions of visitors each year.

Today, Rome is the capital city of Italy and remains a vibrant center of culture, history, and tourism. It is a UNESCO World Heritage Site and continues to be a symbol of the enduring legacy of the Roman Empire."}]

Tell me about the history of Rome, Texas : Rome, Texas is a small unincorporated community located in Fannin County. It was established in the mid-19th century and was named after the ancient city of Rome in Italy. The community grew as a trading post and agricultural center, serving local farmers and ranchers. Today, Rome is known for its rural charm and historic buildings, reflecting its rich history as a frontier settlement in Texas.

Tell me about the history of Rome, Italy : Rome, Italy, is a city with a rich and extensive history that dates back over 2,800 years. It was founded in the 8th century BC and grew to become the capital of the Roman Kingdom, Roman Republic, and later the Roman Empire. Rome played a crucial role in the development of Western civilization, serving as the center of politics, culture, and military power for centuries.

The city of Rome is known for its iconic landmarks, such as the Colosseum, the Roman Forum, and the Pantheon, which stand as a testament to its ancient past. Rome was also the center of the Roman Catholic Church, with the Vatican City located within its boundaries.

Throughout its history, Rome has faced numerous invasions, sackings, and changes in power. The fall of the Western Roman Empire in 476 AD marked the beginning of the Middle Ages in Europe, but Rome continued to be a significant city, albeit with reduced influence.

In the Renaissance period, Rome experienced a revival of arts and culture under the patronage of the papacy, leading to the creation of masterpieces by artists such as Michelangelo, Raphael, and Bernini. The city's architectural and artistic heritage from this period continues to attract millions of visitors each year.

Today, Rome is the capital city of Italy and remains a vibrant center of culture, history, and tourism. It is a UNESCO World Heritage Site and continues to be a symbol of the enduring legacy of the Roman Empire.

One story takes 2 days, and there are 20 stories : The 20 stories will take 40 days to complete.

Action Planning Agent

In [21]: `%run ./action_planning_agent.py`

1. Crack eggs into a bowl
2. Beat eggs with a fork until mixed
3. Heat pan with butter or oil over medium heat
4. Pour egg mixture into pan
5. Stir gently as eggs cook
6. Remove from heat when eggs are just set but still moist
7. Season with salt and pepper
8. Serve immediately

Results

All the output of the test script is SUCCESS.